Lab 2 – Product Description

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Table of Contents

1 Introduction 2
1.1 Purpose 2
1.2 Scope 2
1.3 Definitions, Acronyms, and Abbreviations 3
1.4 References 3
1.5 Overview 5
2 Overall Description 5
2.1 Product Perspective 5
2.2 Product Functions 6
2.3 User Characteristics 7
2.4 Constraints 8
2.5 Assumptions and Dependencies 8

List of Figures

Figure 1 5

List of Tables

Table 1 6

1 Introduction

College is supposed to be a time of self-improvement and learning, and while it can sometimes be difficult and stressful, whether that is because of a heavy course load, difficult material, or adjusting to the rigors of college life, there is no need to make things unnecessarily difficult for students who are trying to sign up for their new semester. One of the main problems with online scheduling for college classes seems to stem consistently from the way that the current applications handle scheduling and conflicts. College students and their advisors can often spend overly excessive amounts of time trying to figure out timing conflicts when preparing to register for classes. Required classes may sometimes overlap with each other or students could have other commitments that make it even more challenging to successfully schedule needed courses to finish their degrees. GradMap is a new application that aims at trying to correct these problems and bridge the gap left by traditional scheduling applications.

1.1 Purpose

The GradMap prototype uses a tailored UI that responds to user input to accurately and efficiently assist students and advisors with creating the best possible schedule each semester in order to ensure that course requirements are met as quickly and easily as possible. GradMap will tailor scheduling options to the student's major degree requirements, the pre-requisites that will best help them to further understand the course material in the required courses, the courses that are offered in a given semester, and student time restraints because of conflicting schedules.

GradMap will implement a system that will make scheduling a breeze and reduce stress on the student. The long-term goal of this application is to help make sure that more students successfully complete their degree and enter the community after graduating with all of the tools and knowledge that they need to be successful in their field, without compromising their pocket book or school/life balance.

GradMap is an online application that assists students and advisors with the creation of a course plan. It will not sign the student up for classes. The student will have to use their university's normal sign-up service to register for classes. Advisors will be able to view and give suggestions on the proposed plans created by the student and GradMap, allowing the advisor to accept/reject or comment suggested changes to a plan. The advisor will not be able to use GradMap to block or accept schedules as GradMap is to only be used for the purpose of schedule planning, not implementation.

1.2 Scope

GradMap's prototype will allow students and advisors to easily create plans for new semesters, simplifying the process of sorting through degree requirements much simpler. This process will require most of the Key Features that will be necessary for the final product.

The prototype will allow students to successfully navigate their degree's requirements and will highlight critical courses that must be taken in the academic term listed in order to graduate on time. The program will also bring attention to any hidden prerequisite courses which may

prove beneficial. The auto-building features of GradMap will allow students to create a draft of a class schedule using pre-inputted degree requirements, current course availability, and designated student time restraints. The prototype will also use advanced conflict detection systems to detect overlapping class times, problems with pre-requisites, and excessive credit load in real time so that students and advisors instantly know of any conflicts that may require resolution prior to registration. Advisors will be able to assess the student plans for approval or rejection and offer feedback to the students.

1.3 Definitions, Acronyms, and Abbreviations

Apache: An open-source web server used to deliver web content.

Authenticate User System: A way to provide security to login information.

Auto-building: A program that runs and builds a schedule immediately upon initiation.

Database: A structured collection of data stored electronically.

Java: An object-oriented programming language.

Major Functional Component Diagram (MFCD): A diagram that displays the major components involved in an application.

MySQL: An open-source relational database management system.

Nginx: A high-performance web server and reverse proxy.

Server-side Language: A programming language that executes on a server rather than a client device.

User Interface: The part of a program that a user sees.

Web-based Program: Software accessed through a web browser rather than installed locally.

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1.5 Overview

This SRS provides features and capabilities of the GradMap prototype, as well as the basic hardware and software features of the application. The next section will provide more details about GradMap's freatures.

2 Overall Description

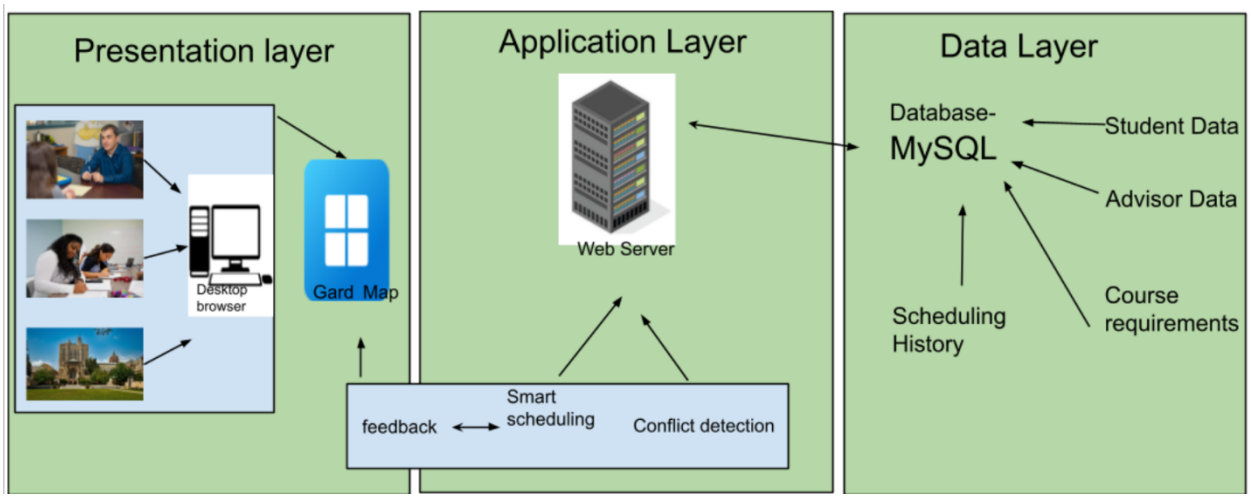
The GradMap prototype will be a web application that will be accessible to students and advisors on any device with internet access. Most features of the completed product will be present in the prototype, though some features may have reduced functionality.

2.1 Product Perspective

GradMap will be an online system that can be accessed on any PC or smartphone. It will make use of the major operating systems, including Windows, MacOS, and Linux, and various web servers such as Apache, Nginx, and Microsoft IIS, making it accessible and able to be used by any University's favored platforms. The program will focus most on the use of Java for the server-side language used to create the website and GradMap will make use of MySQL as the major database storing information such as schedules, course requirements, and user data. The way that GradMap is set up means that, according to the MFCD (as seen in Figure 1), GradMap's actual appearance on the web browser is the Presentation Layer, or what everyone sees when they click on the link. The Application Layer consists of the server being used for GradMap, which handles the smart scheduling aspects of the application, as well as the ability to detect conflicts in schedules. The final layer, the Data layer, includes the database that runs in the background, which holds all of the software parts of the application, including user data, course requirements, and scheduling history to help make sure that the student is staying on track to complete their degree and graduate on time.

Figure 1

GradMap Major Functional Component Diagram



GradMap will include a variety of software in order to enable all of the tasks that will be required of the site. First will be an authenticate user system which will handle the secure login information and provide access to accounts for both students and advisors. Next, GradMap will be capable of auto-building a draft of a class schedule using pre-inputted degree requirements, current course availability, and designated student time restraints while also using advanced conflict detection systems to detect overlapping class times, problems with pre-requisites, and excessive credit load in real time so that students and advisors instantly know of any conflicts that may require resolution prior to registration. In addition, the system will be able to make sure that the proposed plan is degree-compliant by conducting a type of "Fact-checking" against the degree requirements to ensure that the student is staying on track and completing the courses that they need to graduate. Next, the system will check any time conflicts and offer the student and advisor alternative course suggestions that will still help the student move towards their expected graduation without having to guess at what courses are useful for their degree and taking the risk that they may then have to find ways to potentially make an extra semester doable just because they picked an alternate elective that does not help towards the completion of their degree. Finally, the system will send the proposed schedule to the advisor, who will then assess the plan for approval or rejection while the system records feedback to then be sent to the student. Once the student receives feedback via email, they can then either sign up for classes or try to rearrange their schedule to try for approval once again.

2.2 Product Functions

Most of the features that will be included in the real-world product will be present in the prototype, if only in a slightly reduced state of functionality, as seen in Table 1. For students, these features include plan creation, rule checking for degree requirements, simple suggestions for alternatives, detection of time conflicts, and a basic understanding of student history. Advisors will be able to use the advisor dashboard to approve prospective plans, and both Advisors and students will be able to have separate log-in systems and the ability to communicate and collaborate on the plan-creating process.

Table 1

Table 1 of Comparison Between RWP and Prototype

Features	Real-World Product (RWP)	Prototype
Automated Degree Requirement Checker	Automatically updates degree requirements across different programs.	Basic rule-checking with manually entered program requirements.

Auto-Plan Builder	Generates complete, conflict-free academic plans.	Creates simplified schedules using a limited set of restraints.
Smart Suggestions	Delivers course alternatives, elective recommendations, and optimized pathways.	Simple rule-based suggestions to highlight the idea without full AI.
Student Profile Integration	Fully integrates with the university system to import data. (Coursework, degree progress, etc.)	Uses manual input for student data due to lack of SIS access.
Advisor Dashboard	Provides advisors with review tools, analytics, and intervention insights.	Offers a basic approval and commenting interface.
Real-Time Conflict Detection	Instantly flags time conflicts, prerequisite issues, and policy issues.	Detects only time conflicts and basic prerequisites.
Collaboration Tools	Supports real-time communication and plan editing between students and advisors.	A simple comment system without real-time syncing or messaging.
Authentication and Role Management	Provides secure logins for students, advisors, and administrators.	Basic separate login pages to differentiate student and advisor access.

These features will be partially implemented, allowing basic use of the system to showcase the understanding of how the UI and AI should work together to create the best possible plan suggestions. Students will be able to enter search parameters to find courses, see the courses that are currently available, and input their time constraints in order to filter for the courses that will fit best into their existing schedule. The student will also be able to enter their current course history and degree in order to filter out courses that they have already taken towards their degree, making it easier to find the classes that they still need to complete. Advisors will be able to view the plans of their students and make comments on the changes that they would suggest as well as be able to approve or reject plans that the student presents to them for approval prior to course registration.

2.3 User Characteristics

The user roles that are expected to be present in the use of the prototype are that of the student, advisor, and administrator. Students are expected to be at the lowest user expertise level, and the prototype should allow clear understanding of how the program functions while also

giving the student the ability to contact advisors or even admins for assistance. The prototype should not allow the student to finalize their schedule plan without the approval of an advisor. Advisors should be at the middle expertise level, capable of viewing students plans for approval and assisting students with understanding how the program works while being capable of walking the students through the use of the program if necessary. The advisors should not be capable of making any changes to student plans or degree requirements that have been entered into the program. They should only be able to approve, deny, or suggest changes to student plans. Administrators should be at the highest expertise level. They should be capable of entering degree requirements, available courses, and granting access to the program for students and advisors. They should not be able to view or make changes to any student plans.

2.4 Constraints

As with any internet service, there is always the risk for data breaches or theft of personal information.

2.5 Assumptions and Dependencies

The prototype will require access to university databases in order to retrieve course and degree requirements and availability.